

CareerMatch AI – AI/ML Skills Knowledge Guide

Generalized Educational Knowledge – Hackathon Prototype

Introduction

This document is a structured reference of core AI/ML-related skills used by CareerMatch AI to compare a student's resume against a target job role and identify skill gaps. Each skill below includes ten structured fields so the retrieval system can consistently answer questions about what a skill is, why it matters, and how it fits into a learning path. This is generalized educational content for a hackathon prototype and is not tied to any employer.

Skill Reference (A–Z by learning relevance)

Python

Category: Programming Language

Simple Definition: A general-purpose, beginner-friendly programming language widely used for data work, automation, and AI development.

Why It Matters: It is the primary language for machine learning, data analysis, and AI tooling because of its simple syntax and large ecosystem of libraries.

Prerequisites: Basic computer literacy; no prior coding experience required.

Beginner Knowledge: Can write scripts using variables, loops, conditionals, functions, and lists/dictionaries.

Intermediate Knowledge: Comfortable with object-oriented programming, virtual environments, error handling, and using third-party libraries.

Common Job Roles: Machine Learning Engineer, Data Scientist, Data Analyst, AI Engineer, Python Developer, Software Engineer

Related Skills: NumPy, Pandas, OOP, APIs

Typical Project Usage: Used to load data, build models, write automation scripts, and connect components of an AI pipeline.

Priority: High

SQL

Category: Data / Databases

Simple Definition: A query language used to read, filter, and aggregate data stored in relational databases.

Why It Matters: Most real-world data lives in databases, so extracting and shaping it correctly is a foundational skill for any data-related role.

Prerequisites: Basic understanding of tables and spreadsheets.

Beginner Knowledge: Can write SELECT, WHERE, GROUP BY, and simple JOIN queries.

Intermediate Knowledge: Comfortable with subqueries, window functions, and query optimization.

Common Job Roles: Data Analyst, Data Scientist, Data Engineer, Machine Learning Engineer

Related Skills: Databases, PostgreSQL, MySQL, Data Cleaning

Typical Project Usage: Used to pull structured datasets from a company database before analysis or model training.

Priority: High

C++

Category: Programming Language

Simple Definition: A compiled, performance-oriented programming language used for systems and high-performance applications.

Why It Matters: It is valued for roles where speed and memory control matter, such as backend systems and competitive programming.

Prerequisites: Basic programming logic.

Beginner Knowledge: Understands variables, loops, functions, and basic pointers.

Intermediate Knowledge: Comfortable with object-oriented design, memory management, and standard template library.

Common Job Roles: Software Engineer

Related Skills: Data Structures, Algorithms

Typical Project Usage: Used in coding interviews and performance-critical backend components.

Priority: Medium

Java

Category: Programming Language

Simple Definition: An object-oriented programming language widely used for enterprise backend applications.

Why It Matters: Many large organizations run backend systems in Java, making it a common requirement for software roles.

Prerequisites: Basic programming logic.

Beginner Knowledge: Can write simple classes, loops, and conditionals.

Intermediate Knowledge: Comfortable with OOP design patterns and basic frameworks.

Common Job Roles: Software Engineer

Related Skills: OOP, Data Structures

Typical Project Usage: Used to build backend services and enterprise applications.

Priority: Medium

NumPy

Category: Library

Simple Definition: A Python library for fast numerical computation using arrays and matrices.

Why It Matters: It underlies almost every data science and machine learning library in Python.

Prerequisites: Basic Python.

Beginner Knowledge: Can create arrays and perform basic mathematical operations on them.

Intermediate Knowledge: Comfortable with broadcasting, vectorized operations, and linear algebra functions.

Common Job Roles: Machine Learning Engineer, Data Scientist, AI/ML Intern

Related Skills: Python, Pandas, Linear Algebra

Typical Project Usage: Used for numerical transformations before feeding data into a model.

Priority: High

Pandas

Category: Library

Simple Definition: A Python library for loading, cleaning, and manipulating tabular data.

Why It Matters: It is the standard tool for everyday data wrangling tasks before analysis or modeling.

Prerequisites: Basic Python, basic NumPy.

Beginner Knowledge: Can load a CSV, filter rows, and compute simple summary statistics.

Intermediate Knowledge: Comfortable with merging datasets, handling missing values, and group-by operations.

Common Job Roles: Data Scientist, Data Analyst, Machine Learning Engineer, AI/ML Intern

Related Skills: NumPy, Data Cleaning, EDA

Typical Project Usage: Used to clean and reshape a dataset before exploratory analysis or model training.

Priority: High

Matplotlib

Category: Library

Simple Definition: A Python plotting library used to create charts and graphs.

Why It Matters: Visualizing data helps identify trends, outliers, and relationships that raw numbers hide.

Prerequisites: Basic Python, basic Pandas.

Beginner Knowledge: Can create simple line, bar, and scatter plots.

Intermediate Knowledge: Comfortable customizing plots and building multi-panel figures.

Common Job Roles: Data Scientist, Data Analyst

Related Skills: Data Visualization, Pandas

Typical Project Usage: Used to visualize distributions and trends during exploratory data analysis.

Priority: Medium

Statistics

Category: Mathematics

Simple Definition: The study of collecting, analyzing, and interpreting numerical data.

Why It Matters: Machine learning is built on statistical reasoning, from describing data to validating model results.

Prerequisites: Basic mathematics.

Beginner Knowledge: Understands mean, median, mode, standard deviation, and basic distributions.

Intermediate Knowledge: Comfortable with hypothesis testing, confidence intervals, and correlation analysis.

Common Job Roles: Data Scientist, Data Analyst, Machine Learning Engineer

Related Skills: Probability, Data Visualization

Typical Project Usage: Used to summarize data and validate whether observed patterns are meaningful.

Priority: High

Probability

Category: Mathematics

Simple Definition: The branch of mathematics that studies uncertainty and the likelihood of events.

Why It Matters: Many machine learning algorithms are built directly on probabilistic reasoning.

Prerequisites: Basic statistics.

Beginner Knowledge: Understands basic probability rules and simple distributions.

Intermediate Knowledge: Comfortable with conditional probability, Bayes theorem, and common distributions.

Common Job Roles: Data Scientist, Machine Learning Engineer

Related Skills: Statistics, Machine Learning

Typical Project Usage: Used to reason about model confidence and uncertainty in predictions.

Priority: Medium

Linear Algebra

Category: Mathematics

Simple Definition: The branch of mathematics dealing with vectors, matrices, and linear transformations.

Why It Matters: It is the mathematical backbone of machine learning and deep learning algorithms.

Prerequisites: Basic mathematics.

Beginner Knowledge: Understands vectors, matrices, and basic matrix operations.

Intermediate Knowledge: Comfortable with eigenvalues, matrix decomposition, and their role in ML algorithms.

Common Job Roles: Machine Learning Engineer, Data Scientist

Related Skills: NumPy, Statistics

Typical Project Usage: Used conceptually to understand how models like neural networks transform data.

Priority: Medium

Data Cleaning

Category: Data Processing

Simple Definition: The process of fixing or removing incorrect, incomplete, or inconsistent data.

Why It Matters: Poor quality data leads to unreliable models, regardless of how advanced the algorithm is.

Prerequisites: Basic Pandas.

Beginner Knowledge: Can handle missing values, duplicates, and obvious formatting errors.

Intermediate Knowledge: Comfortable designing repeatable cleaning pipelines for messy real-world data.

Common Job Roles: Data Analyst, Data Scientist, Machine Learning Engineer

Related Skills: Pandas, EDA

Typical Project Usage: Used as the first practical step in nearly every data project.

Priority: High

EDA

Category: Data Processing

Simple Definition: Exploratory Data Analysis: examining a dataset to understand its structure, patterns, and issues before modeling.

Why It Matters: It reveals patterns, anomalies, and relationships that guide feature engineering and model choice.

Prerequisites: Pandas, basic statistics.

Beginner Knowledge: Can summarize a dataset and create basic visualizations.

Intermediate Knowledge: Comfortable identifying correlations, outliers, and data quality issues systematically.

Common Job Roles: Data Scientist, Data Analyst

Related Skills: Pandas, Matplotlib, Statistics

Typical Project Usage: Used at the start of a project to decide which features and models are worth pursuing.

Priority: High

Feature Engineering

Category: Machine Learning

Simple Definition: The process of creating or transforming input variables to improve model performance.

Why It Matters: Well-designed features often matter more to model accuracy than the choice of algorithm.

Prerequisites: EDA, basic statistics.

Beginner Knowledge: Can create simple derived columns and encode categorical variables.

Intermediate Knowledge: Comfortable with scaling, interaction features, and domain-driven feature design.

Common Job Roles: Machine Learning Engineer, Data Scientist

Related Skills: EDA, Scikit-learn

Typical Project Usage: Used to convert raw data into a form that a model can learn from effectively.

Priority: High

Scikit-learn

Category: Library

Simple Definition: A Python library that provides ready-to-use machine learning algorithms and evaluation tools.

Why It Matters: It allows learners to implement and test standard ML models without writing algorithms from scratch.

Prerequisites: Python, NumPy, Pandas.

Beginner Knowledge: Can train a simple classifier or regressor using the library's standard interface.

Intermediate Knowledge: Comfortable with pipelines, cross-validation, and hyperparameter tuning.

Common Job Roles: Machine Learning Engineer, Data Scientist, AI/ML Intern

Related Skills: Machine Learning, Model Evaluation

Typical Project Usage: Used to quickly prototype and evaluate classical machine learning models.

Priority: High

Machine Learning

Category: Core Concept

Simple Definition: A field of AI where systems learn patterns from data rather than being explicitly programmed.

Why It Matters: It is the core discipline behind most modern AI products, from recommendations to predictions.

Prerequisites: Python, statistics, basic linear algebra.

Beginner Knowledge: Understands the difference between supervised and unsupervised learning and can train a basic model.

Intermediate Knowledge: Comfortable choosing algorithms, tuning models, and interpreting evaluation metrics.

Common Job Roles: Machine Learning Engineer, Data Scientist, AI Engineer

Related Skills: Scikit-learn, Feature Engineering, Model Evaluation

Typical Project Usage: Used to build predictive systems from historical data.

Priority: High

Regression

Category: Machine Learning

Simple Definition: A supervised learning technique used to predict continuous numerical values.

Why It Matters: Many real-world problems, like predicting prices or demand, require numeric predictions.

Prerequisites: Statistics, Scikit-learn.

Beginner Knowledge: Can train and interpret a simple linear regression model.

Intermediate Knowledge: Comfortable with regularized regression and evaluating fit using residuals.

Common Job Roles: Data Scientist, Machine Learning Engineer

Related Skills: Machine Learning, Model Evaluation

Typical Project Usage: Used in projects that forecast quantities such as sales or resource demand.

Priority: Medium

Classification

Category: Machine Learning

Simple Definition: A supervised learning technique used to predict a category or label.

Why It Matters: Many practical AI problems, such as spam detection, are classification problems.

Prerequisites: Statistics, Scikit-learn.

Beginner Knowledge: Can train a basic classifier and interpret accuracy.

Intermediate Knowledge: Comfortable comparing multiple classifiers and handling imbalanced classes.

Common Job Roles: Data Scientist, Machine Learning Engineer

Related Skills: Machine Learning, Model Evaluation

Typical Project Usage: Used in projects that sort items into predefined groups.

Priority: High

Clustering

Category: Machine Learning

Simple Definition: An unsupervised learning technique that groups similar data points without predefined labels.

Why It Matters: It helps discover hidden structure in data when labels are not available.

Prerequisites: Statistics, Scikit-learn.

Beginner Knowledge: Can apply a simple clustering algorithm and visualize the groups.

Intermediate Knowledge: Comfortable choosing the number of clusters and validating cluster quality.

Common Job Roles: Data Scientist

Related Skills: Machine Learning, EDA

Typical Project Usage: Used in projects like customer segmentation.

Priority: Medium

Model Evaluation

Category: Machine Learning

Simple Definition: The process of measuring how well a trained model performs.

Why It Matters: A model is only useful if its performance is measured correctly and honestly.

Prerequisites: Machine Learning basics.

Beginner Knowledge: Can calculate accuracy and understand a train/test split.

Intermediate Knowledge: Comfortable with precision, recall, F1 score, confusion matrices, and cross-validation.

Common Job Roles: Machine Learning Engineer, Data Scientist

Related Skills: Machine Learning, Statistics

Typical Project Usage: Used to compare candidate models and select the best one for deployment.

Priority: High

Deep Learning

Category: Machine Learning

Simple Definition: A subset of machine learning that uses multi-layered neural networks to learn complex patterns.

Why It Matters: It powers most modern breakthroughs in vision, language, and speech-related AI.

Prerequisites: Machine Learning, linear algebra.

Beginner Knowledge: Understands what a neural network is and can train a simple one on a small dataset.

Intermediate Knowledge: Comfortable designing network architectures and tuning training parameters.

Common Job Roles: AI Engineer, Machine Learning Engineer, NLP Engineer, Computer Vision Engineer

Related Skills: TensorFlow, PyTorch, CNN

Typical Project Usage: Used for tasks that involve images, text, or audio where patterns are complex.

Priority: High

TensorFlow

Category: Framework

Simple Definition: An open-source deep learning framework developed for building and training neural networks.

Why It Matters: It is one of the two dominant frameworks used in industry for deep learning development.

Prerequisites: Python, deep learning basics.

Beginner Knowledge: Can build and train a simple neural network using high-level APIs.

Intermediate Knowledge: Comfortable customizing architectures and training loops.

Common Job Roles: AI Engineer, Machine Learning Engineer, Computer Vision Engineer

Related Skills: Deep Learning, PyTorch

Typical Project Usage: Used to build and deploy neural network models.

Priority: Medium

PyTorch

Category: Framework

Simple Definition: An open-source deep learning framework known for flexibility and ease of experimentation.

Why It Matters: It is widely used in research and increasingly in production AI systems.

Prerequisites: Python, deep learning basics.

Beginner Knowledge: Can build and train a simple neural network using standard modules.

Intermediate Knowledge: Comfortable with custom layers, training loops, and GPU usage.

Common Job Roles: AI Engineer, Machine Learning Engineer, NLP Engineer, Computer Vision Engineer

Related Skills: Deep Learning, TensorFlow

Typical Project Usage: Used to build, train, and fine-tune neural network models.

Priority: Medium

CNN

Category: Deep Learning Concept

Simple Definition: Convolutional Neural Network: a deep learning architecture specialized for processing images.

Why It Matters: It is the foundation of most computer vision systems, from classification to detection.

Prerequisites: Deep Learning basics.

Beginner Knowledge: Understands what convolution and pooling layers do at a conceptual level.

Intermediate Knowledge: Comfortable building and fine-tuning CNN architectures for real datasets.

Common Job Roles: Computer Vision Engineer, AI Engineer

Related Skills: Deep Learning, Image Classification, Transfer Learning

Typical Project Usage: Used to classify or detect objects within images.

Priority: High

NLP

Category: Core Concept

Simple Definition: Natural Language Processing: the field of AI focused on understanding and generating human language.

Why It Matters: It powers chatbots, search engines, sentiment analysis, and modern generative AI systems.

Prerequisites: Python, Machine Learning basics.

Beginner Knowledge: Understands tokenization and can build a simple text classifier.

Intermediate Knowledge: Comfortable using pretrained language models for real tasks.

Common Job Roles: NLP Engineer, AI Engineer, Data Scientist

Related Skills: Transformers, Embeddings, Text Classification

Typical Project Usage: Used to process, classify, or generate text-based content.

Priority: High

Transformers

Category: Deep Learning Architecture

Simple Definition: A neural network architecture that processes sequences using an attention mechanism, forming the basis of modern language models.

Why It Matters: Nearly all state-of-the-art NLP and generative AI systems are built on transformer architecture.

Prerequisites: Deep Learning, NLP basics.

Beginner Knowledge: Understands the general idea of attention and why transformers replaced older sequence models.

Intermediate Knowledge: Comfortable fine-tuning pretrained transformer models for specific tasks.

Common Job Roles: NLP Engineer, AI Engineer

Related Skills: BERT, LLMs, NLP

Typical Project Usage: Used as the underlying architecture for language understanding and generation tasks.

Priority: High

BERT

Category: Model

Simple Definition: A pretrained transformer-based language model designed for understanding text context in both directions.

Why It Matters: It set a strong baseline for many NLP tasks and remains widely used for text understanding.

Prerequisites: Transformers, NLP basics.

Beginner Knowledge: Understands what BERT is used for and can use it for a simple classification task.

Intermediate Knowledge: Comfortable fine-tuning BERT-family models for domain-specific problems.

Common Job Roles: NLP Engineer

Related Skills: Transformers, NLP, Embeddings

Typical Project Usage: Used for tasks such as sentiment analysis, text classification, and question answering.

Priority: Medium

LLMs

Category: Core Concept

Simple Definition: Large Language Models: very large transformer-based models trained on massive text data to understand and generate language.

Why It Matters: They are the foundation of most current generative AI applications, from chatbots to coding assistants.

Prerequisites: Transformers, NLP basics.

Beginner Knowledge: Understands what an LLM is and how to interact with one through an API or interface.

Intermediate Knowledge: Comfortable designing prompts and integrating LLMs into applications.

Common Job Roles: AI Engineer, NLP Engineer

Related Skills: Prompt Engineering, RAG, Transformers

Typical Project Usage: Used to power chatbots, content generation, and reasoning-based applications.

Priority: High

Generative AI

Category: Core Concept

Simple Definition: A category of AI systems that create new content such as text, images, or code, rather than only analyzing data.

Why It Matters: It has become one of the fastest-growing areas of AI with wide industry adoption.

Prerequisites: Machine Learning, NLP basics.

Beginner Knowledge: Understands common use cases and can use existing generative tools effectively.

Intermediate Knowledge: Comfortable building applications that combine generative models with custom logic.

Common Job Roles: AI Engineer

Related Skills: LLMs, Prompt Engineering, RAG

Typical Project Usage: Used to build applications like content generators, assistants, and creative tools.

Priority: High

Prompt Engineering

Category: Practical Skill

Simple Definition: The practice of designing effective instructions to get useful outputs from a language model.

Why It Matters: Well-designed prompts significantly improve the reliability and quality of AI-generated outputs.

Prerequisites: Basic understanding of LLMs.

Beginner Knowledge: Can write clear, specific prompts and iterate based on results.

Intermediate Knowledge: Comfortable using structured prompting techniques for complex, multi-step tasks.

Common Job Roles: AI Engineer, NLP Engineer

Related Skills: LLMs, RAG

Typical Project Usage: Used to control the behavior and output format of a generative AI application.

Priority: Medium

Embeddings

Category: Core Concept

Simple Definition: Numerical vector representations of text, images, or other data that capture semantic meaning.

Why It Matters: They allow computers to measure similarity between pieces of content, powering search and retrieval systems.

Prerequisites: Basic Machine Learning.

Beginner Knowledge: Understands that embeddings turn data into vectors that capture meaning.

Intermediate Knowledge: Comfortable generating and comparing embeddings for a retrieval-based application.

Common Job Roles: AI Engineer, NLP Engineer

Related Skills: Vector Databases, RAG, NLP

Typical Project Usage: Used to find semantically similar content, such as matching a resume to a job description.

Priority: High

RAG

Category: Core Concept

Simple Definition: Retrieval-Augmented Generation: a technique that retrieves relevant information from a knowledge source before generating an AI response.

Why It Matters: It grounds AI responses in real, retrievable data instead of relying only on a model's internal memory.

Prerequisites: Embeddings, Vector Databases, LLM basics.

Beginner Knowledge: Understands the retrieve-then-generate pipeline at a conceptual level.

Intermediate Knowledge: Comfortable building an end-to-end RAG pipeline with a vector database.

Common Job Roles: AI Engineer, NLP Engineer

Related Skills: Embeddings, Vector Databases, LLMs

Typical Project Usage: Used to build systems, like this hackathon project, that answer questions using a custom knowledge base.

Priority: High

LangChain

Category: Framework

Simple Definition: A framework that simplifies building applications powered by language models, including RAG pipelines.

Why It Matters: It reduces the boilerplate needed to connect LLMs, data sources, and retrieval systems.

Prerequisites: Python, basic LLM concepts.

Beginner Knowledge: Can use pre-built components to construct a simple chain or retrieval pipeline.

Intermediate Knowledge: Comfortable customizing chains, agents, and memory for complex applications.

Common Job Roles: AI Engineer

Related Skills: RAG, LLMs, Vector Databases

Typical Project Usage: Used to orchestrate the retrieval and generation steps of an AI application.

Priority: Medium

LlamaIndex

Category: Framework

Simple Definition: A data framework that helps connect custom data sources to language models for retrieval-based applications.

Why It Matters: It simplifies indexing and querying documents for RAG-style systems.

Prerequisites: Python, basic LLM concepts.

Beginner Knowledge: Can index a set of documents and query them using a simple pipeline.

Intermediate Knowledge: Comfortable customizing indexing and retrieval strategies.

Common Job Roles: AI Engineer

Related Skills: RAG, Vector Databases

Typical Project Usage: Used to build the document indexing layer of a retrieval-based application.

Priority: Medium

FAISS

Category: Library

Simple Definition: A library for efficient similarity search over large sets of vector embeddings.

Why It Matters: It enables fast retrieval of the most relevant content in large-scale RAG systems.

Prerequisites: Embeddings basics.

Beginner Knowledge: Can create a simple vector index and perform a nearest-neighbor search.

Intermediate Knowledge: Comfortable tuning index types for speed and accuracy trade-offs.

Common Job Roles: AI Engineer

Related Skills: Vector Databases, Embeddings, RAG

Typical Project Usage: Used as the retrieval engine that finds relevant chunks of a knowledge base.

Priority: Medium

Vector Databases

Category: Infrastructure

Simple Definition: Specialized databases designed to store and search vector embeddings efficiently.

Why It Matters: They make it possible to search large knowledge bases by meaning rather than exact keyword match.

Prerequisites: Embeddings basics.

Beginner Knowledge: Understands what a vector database stores and why it differs from a normal database.

Intermediate Knowledge: Comfortable setting up and querying a vector database for a retrieval application.

Common Job Roles: AI Engineer, NLP Engineer

Related Skills: Embeddings, FAISS, RAG

Typical Project Usage: Used to store document chunks so relevant ones can be retrieved quickly during a query.

Priority: High

Git

Category: Tooling

Simple Definition: A version control system used to track changes in code over time.

Why It Matters: It is essential for collaborating on code and maintaining a reliable project history.

Prerequisites: Basic command-line familiarity.

Beginner Knowledge: Can commit changes, create branches, and push to a remote repository.

Intermediate Knowledge: Comfortable resolving merge conflicts and using pull requests in a team workflow.

Common Job Roles: Software Engineer, Python Developer, Machine Learning Engineer

Related Skills: GitHub

Typical Project Usage: Used to manage and share project code, including hackathon submissions.

Priority: Medium

Docker

Category: Tooling

Simple Definition: A platform for packaging applications and their dependencies into portable containers.

Why It Matters: It ensures an application runs consistently across different machines and environments.

Prerequisites: Basic command-line familiarity.

Beginner Knowledge: Can write a simple Dockerfile and run a container.

Intermediate Knowledge: Comfortable using multi-container setups and deploying containerized applications.

Common Job Roles: Machine Learning Engineer, AI Engineer, Data Engineer, Python Developer

Related Skills: Cloud Computing, MLOps

Typical Project Usage: Used to package a model or application so it can be deployed reliably.

Priority: Medium

FastAPI

Category: Framework

Simple Definition: A modern Python web framework used to build APIs quickly.

Why It Matters: It is a common choice for exposing machine learning models or backend logic as a web service.

Prerequisites: Python basics.

Beginner Knowledge: Can build a simple API endpoint that returns data.

Intermediate Knowledge: Comfortable building structured APIs with validation and documentation.

Common Job Roles: Python Developer, Machine Learning Engineer, AI Engineer

Related Skills: APIs, Model Deployment

Typical Project Usage: Used to serve a trained model or application logic over the web.

Priority: Medium

Streamlit

Category: Framework

Simple Definition: A Python framework for quickly building simple, interactive web interfaces.

Why It Matters: It allows developers to demo AI and data projects without needing frontend expertise.

Prerequisites: Python basics.

Beginner Knowledge: Can build a simple interactive page to display data or model predictions.

Intermediate Knowledge: Comfortable building multi-page apps with interactive widgets.

Common Job Roles: Data Scientist, AI/ML Intern, Machine Learning Engineer

Related Skills: Model Deployment, Python

Typical Project Usage: Used to create a demo interface for a hackathon or portfolio project.

Priority: Medium

REST APIs

Category: Core Concept

Simple Definition: A common architectural style for building web services that communicate over HTTP.

Why It Matters: Most modern applications, including AI systems, communicate through APIs.

Prerequisites: Basic programming.

Beginner Knowledge: Understands GET/POST requests and can call a simple API.

Intermediate Knowledge: Comfortable designing and securing APIs for real applications.

Common Job Roles: Python Developer, Software Engineer, AI Engineer

Related Skills: FastAPI, Flask

Typical Project Usage: Used to connect a model or backend service to a frontend or other application.

Priority: Medium

Cloud Computing

Category: Infrastructure

Simple Definition: The delivery of computing resources such as storage and servers over the internet.

Why It Matters: Most production AI and data systems run on cloud infrastructure rather than local machines.

Prerequisites: Basic understanding of computing concepts.

Beginner Knowledge: Understands what cloud services are used for at a conceptual level.

Intermediate Knowledge: Comfortable deploying a simple application or model on a cloud platform.

Common Job Roles: Machine Learning Engineer, Data Engineer, AI Engineer

Related Skills: Docker, MLOps

Typical Project Usage: Used to host datasets, models, or applications at scale.

Priority: Medium

MLOps

Category: Practical Skill

Simple Definition: The set of practices for reliably deploying, monitoring, and maintaining machine learning models in production.

Why It Matters: A model that works well in a notebook still needs a reliable path to production, which MLOps provides.

Prerequisites: Machine Learning, Docker, Cloud basics.

Beginner Knowledge: Understands the basic lifecycle of a model from training to deployment.

Intermediate Knowledge: Comfortable setting up monitoring, versioning, and retraining pipelines.

Common Job Roles: Machine Learning Engineer, AI Engineer

Related Skills: Docker, Cloud Computing, Model Deployment

Typical Project Usage: Used to keep deployed models reliable and up to date over time.

Priority: Medium

Model Deployment

Category: Practical Skill

Simple Definition: The process of making a trained model available for use in a real application.

Why It Matters: A model only creates value once it can be accessed and used outside of a notebook.

Prerequisites: Machine Learning basics, basic APIs.

Beginner Knowledge: Can wrap a simple model in an API or interactive app.

Intermediate Knowledge: Comfortable deploying models with proper versioning and monitoring.

Common Job Roles: Machine Learning Engineer, AI Engineer, Computer Vision Engineer

Related Skills: FastAPI, Streamlit, Docker, MLOps

Typical Project Usage: Used to turn a trained model into a usable product or demo.

Priority: High

Data Visualization

Category: Practical Skill

Simple Definition: The practice of representing data visually through charts and graphs.

Why It Matters: It communicates findings clearly to both technical and non-technical audiences.

Prerequisites: Basic statistics, basic Pandas.

Beginner Knowledge: Can create basic charts to summarize a dataset.

Intermediate Knowledge: Comfortable designing dashboards that tell a clear data story.

Common Job Roles: Data Analyst, Data Scientist

Related Skills: Matplotlib, Power BI, Tableau

Typical Project Usage: Used to present analysis results and support data-driven decisions.

Priority: Medium

ETL

Category: Data Engineering

Simple Definition: Extract, Transform, Load: the process of moving data from source systems into a usable format for analysis.

Why It Matters: Reliable data pipelines are the foundation that all downstream analysis and machine learning depend on.

Prerequisites: SQL, Python basics.

Beginner Knowledge: Understands the ETL process conceptually and can write a simple data pipeline script.

Intermediate Knowledge: Comfortable building scheduled, automated pipelines that handle real-world data issues.

Common Job Roles: Data Engineer

Related Skills: Data Pipelines, SQL, Databases

Typical Project Usage: Used to move and prepare data before it reaches analysts or data scientists.

Priority: High

Big Data

Category: Core Concept

Simple Definition: Datasets and systems that are too large or fast-moving for traditional single-machine processing.

Why It Matters: Many organizations need distributed systems to store and process data at scale.

Prerequisites: Basic databases, basic programming.

Beginner Knowledge: Understands why distributed processing is needed for very large datasets.

Intermediate Knowledge: Comfortable working with distributed processing concepts like Spark.

Common Job Roles: Data Engineer

Related Skills: Spark, ETL, Cloud Computing

Typical Project Usage: Used conceptually when designing systems that handle large-scale data.

Priority: Medium

1. Skill Categories

Programming Languages: Python, SQL, C++, Java

Core Libraries: NumPy, Pandas, Matplotlib, Scikit-learn

Mathematics & Statistics: Statistics, Probability, Linear Algebra

Data Preparation: Data Cleaning, EDA, Feature Engineering, Data Visualization

Classical Machine Learning: Machine Learning, Regression, Classification, Clustering, Model Evaluation

Deep Learning: Deep Learning, TensorFlow, PyTorch, CNN

NLP & Generative AI: NLP, Transformers, BERT, LLMs, Generative AI, Prompt Engineering, Embeddings, RAG

RAG & Retrieval Tooling: LangChain, LlamaIndex, FAISS, Vector Databases

Engineering & Deployment: Git, Docker, FastAPI, Streamlit, REST APIs, Cloud Computing, MLOps, Model Deployment

Data Engineering: ETL, Big Data

2. Beginner-to-Advanced Learning Order

- 1 Python programming fundamentals
- 2 Git for version control
- 3 NumPy and Pandas for data handling
- 4 Statistics, probability, and linear algebra foundations
- 5 SQL for querying structured data
- 6 Data cleaning, EDA, and data visualization
- 7 Feature engineering
- 8 Classical machine learning with Scikit-learn (regression, classification, clustering)
- 9 Model evaluation techniques
- 10 Deep learning fundamentals with TensorFlow or PyTorch
- 11 CNNs for vision or Transformers/NLP for language, based on interest
- 12 Generative AI: prompt engineering, embeddings, and LLMs
- 13 Vector databases and retrieval-augmented generation (RAG)
- 14 APIs and frameworks such as FastAPI and Streamlit
- 15 Docker, cloud basics, and MLOps for deployment

3. Common Skill Combinations for AI/ML Roles

- Python + Pandas + Scikit-learn + Model Evaluation → classical Machine Learning Engineer track
- Python + SQL + Data Cleaning + Data Visualization → Data Analyst track
- Python + Statistics + EDA + Machine Learning → Data Scientist track
- Python + Deep Learning + Transformers + Embeddings → NLP Engineer track
- Python + OpenCV-adjacent CNN skills + Transfer Learning → Computer Vision Engineer track
- Python + LLMs + Embeddings + Vector Databases + RAG → AI Engineer / Generative AI track
- SQL + Data Modeling + ETL + Data Pipelines → Data Engineer track
- Python + APIs + Docker + Cloud Computing + MLOps → Deployment-focused ML/AI Engineer track

4. RAG Retrieval Keywords

AI ML skills guide, skill gap analysis, Python skills, SQL skills, machine learning skills, deep learning skills, generative AI skills, RAG skills, vector database, MLOps, resume skill matching, career skill roadmap

Document Metadata

Document Type	AI/ML Skills Knowledge Guide
Category	Artificial Intelligence / Machine Learning
Skill Count	46
Data Status	Generalized Educational Knowledge